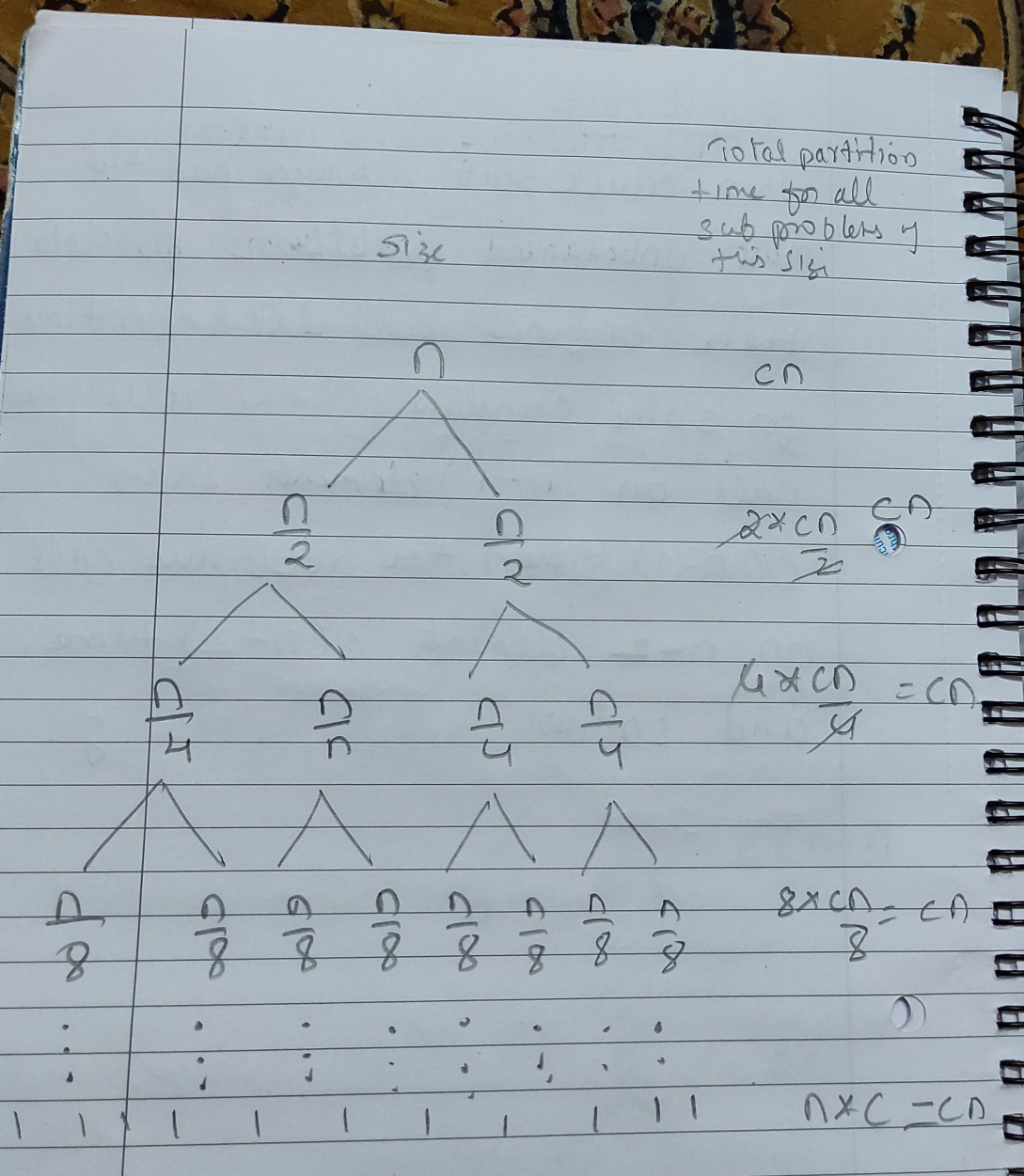
Quick Sort time complexity

Best case:

Best case occurs when the partitions are as evenly balanced as possible their sizes either equal or differ by 1. The former case occurs of the array or subarray has an odd number of elements and the pivot is right in the middle after partitioning and each has $(n-1)/2$ elements. The latter case occurs if the array or subarray has an even number of elements and one partition has $n/2$ elements with other having $n/2 - 1$. In either of these cases, each partition has at most $n/2$ elements.



Total time will summation of them.

so . $h * cn$

$h \rightarrow$ height of a Binary tree

so $h * cn \Rightarrow cn \log_2 n$

using big-Θ notation

we get $O(n \log_2 n)$